

Gaurav Dhingra

Compiler Engineer · LLVM · Systems Programming

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SUMMARY

Compiler engineer with 8 years of software experience and deep expertise in LLVM-based compiler infrastructure. Core contributor to LFortran (LLVM-based Fortran compiler) and active contributor to LLVM/Clang. Background spans compiler IR design, code generation, symbolic computation engines, and expression parsing. Two-time Google Summer of Code participant with a sustained track record of open-source contributions to SymPy, LFortran, and LLVM.

TECHNICAL SKILLS

Languages:	C++, Python, C, Kotlin, Bash
Compiler Infrastructure:	LLVM, Clang, clang-tidy, LLVM IR, ASR (Abstract Semantic Representation)
Tools & Platforms:	Git, CMake, AddressSanitizer (ASan), GDB, Docker, CI/CD (GitHub Actions)
Domains:	Compiler optimization passes, frontend/backend code generation, symbolic computation, expression parsing, intermediate representations

EXPERIENCE

LLVM Project — Open-Source Contributor

2025 – Present

Remote

- Contributing to clang-tidy: developing and refining static analysis checks for the Clang tooling ecosystem
- Investigating and reproducing crash bugs in Clang's parser (e.g., stack-overflow in `ParsePostfixExpressionSuffix`) using locally built LLVM with AddressSanitizer on ARM64
- Engaging with the LLVM community through bug reports, code review, and patch submissions

LFortran — Compiler Engineer

March 2024 – June 2025

Remote

- Core contributor to LFortran, a modern open-source Fortran compiler built on the LLVM backend, working toward its beta release
- Designed and improved the array operation pass in ASR (Abstract Semantic Representation), LFortran's high-level intermediate representation
- Implemented compiler frontend features and optimization passes in C++ using the LLVM C++ API
- Collaborated asynchronously with the open-source team on architecture decisions, code review, and CI workflows

Digital Aristotle (Byju's) — Senior Research Engineer

April 2022 – February 2024

Bangalore, India

- Built a math solver engine for K-12 education: designed and implemented a parser and compiler pipeline that transformed LaTeX mathematical expressions into an AST and internal evaluation form
- Architected a domain-specific SDK in Kotlin for expression tree construction, type analysis, and step-by-step symbolic simplification
- Worked in a cross-functional agile team integrating the solver with GeoGebra's visualization layer

Digital Aristotle (Byju's) — Junior Research Engineer

June 2018 – March 2022

Bangalore, India

- Developed the Autosolver backend for the Byju's BTLA app: a symbolic math engine using SymPy for automated step-by-step problem solving

- Designed and implemented a DSL for declarative step-by-step math solution templates, enabling content authors to define solver behavior without code changes
- Built CI/CD pipelines and test infrastructure in Python and TypeScript

OPEN-SOURCE CONTRIBUTIONS

SymPy — Pull Request Manager (NumFOCUS-funded)

- Selected as one of the top contributors to manage PR review and release workflows for SymPy (12k+ GitHub stars), a widely-used symbolic mathematics library. Position funded by NumFOCUS (Sep 2017, Dec 2017, Feb 2018)

Google Summer of Code 2017 — SymPy

- Extended the Risch integration algorithm: implemented the parametric logarithmic derivative problem and added support for integrating trigonometric functions via the Risch algorithm

Google Summer of Code 2016 — SymPy

- Implemented coset enumeration and Reidemeister–Schreier algorithms for finitely presented groups; added low-index subgroup computation for group-theoretic operations in SymPy

EDUCATION

Indian Institute of Technology Roorkee

2013 – 2018

Integrated M.Sc. in Applied Mathematics

Relevant coursework: Design & Analysis of Algorithms, Data Structures, Graph Theory, Linear Algebra, Discrete Mathematics